Welcome to AI for Creativity

Start here · your getting-started notes for the module

Module	Artificial Intelligence for Creativity
Level · credits	Level 6 · 30 credits
Length	14 weeks · 300 learning hours (90 scheduled + 210 independent)
Assessment	Portfolio · 100% · submitted at the end of Week 14
Lecturer	Dr. Geetha Mohan
Where	Future Education University College, UAE

1· What this module is about

Using open-source machine-learning frameworks (PyTorch, fastai and TensorFlow), you will learn the core principles of working with AI and apply them in Python to create research prototypes for creative purposes. You will train models for computer vision, text, tabular data and recommendations, and consider the ethical implications of your work throughout.

Unit learning objectives

LO1 Demonstrate detailed knowledge of contexts, concepts, technologies, skills, and processes in the creation of a defined project that uses cutting-edge artificial intelligence processes.

LO2 Demonstrate an understanding of a range of programming skills and workflows and present work in a professional manner appropriate to the audience and context.

LO3 Create a body of work that demonstrates a consistent, refined, and ethical application of appropriate advanced programming, and machine learning processes.

The journey in four phases

Weeks	Phase	What you will do
1–4	Foundations	Python through creative coding; data and machine-learning basics; how neural networks learn
5–11	Core AI techniques	Image classification, object detection, text analysis and generation, recommenders, generative image and sound
7	Checkpoint	Pitch your project proposal and get feedback
12–14	Professional practice and realization	Polish your code and interface, test with users, showcase and submit

2 · How each week works

Every week follows the same pattern, and every week on Moodle uses the same headings, so you always know where to look.

When	Session	What happens
Before class	Pre-recorded materials	Watch Videos and Lecture Materials
In class	A · Technical demonstration	New concepts and live coding; type along
In class	B · Lab workshop	Build something creative in pairs (Core or Stretch tasks)
In class	C · Group seminar	Discussion, readings and debate about AI in creative practice
In class	D · Tutorial	Code clinic, feedback and project check-ins
After class	Independent study	Practice notebook, reading and portfolio work

Two tracks in every lab

- **Core** -edit and complete working code. Everyone starts here, and Core work is enough to do well.
- **Stretch** -write new code and extend the idea. Move on to Stretch when you finish Core.

3 · What you need

Item	Details
Laptop	Windows or macOS, charged, at every session. About 5 GB of free space by Week 4.
Google account	For Google Colab, which runs Python in your browser and has nothing to install at first.
Anaconda + JupyterLab	Installed together in Week 4 to work on your own laptop. See the Setup Guide in this section.
R&D journal	A document, sketchbook or blog where you record your work every week. It becomes your R&D document.
Your own material	From Week 2 you will use your own photos, artwork, text or sound. Start collecting now and only use material you have the right to use.

4 · Before Week 1 checklist

- ☐ Log in to Moodle and bookmark this module page
- ☐ Read these starting notes and the 14-week lesson plan
- ☐ Check that you can sign in to Google and open colab.research.google.com
- ☐ Download the Week 1 demo notebook and open it in Colab (Setup Guide, Part 1)
- ☐ Turn on Moodle notifications for Announcements
- ☐ Introduce yourself in the Q&A forum: your name, your discipline, and one thing you hope AI could do for your work
- ☐ Create your R&D journal and add a first entry: 'What I think AI is, before Week 1'
- ☐ Go through Week 1 materials

5 · Assessment at a glance

There is one assessment: a **portfolio worth 100%**, submitted at the end of Week 14. It has three parts:

Part	What it is	Tips
R&D document	Evidence of your learning journey: research, data, experiments, ethics and reflection	Build it every week from your R&D journal. See the annotated exemplar in 'Portfolio & Research Skills'.
AI model	A model that creatively uses processes from the module, with code commented to show its logic	Keep your notebooks tidy and comment as you go.
Essay or video essay	Places your practice within a critical framework	Draft by Week 12 for feedback.

Feedback before you submit — four formative checkpoints (not graded):

Week	Checkpoint
3	R&D journal review
7	Project proposal crit
10	Prototype check-in
12	Draft essay or video-essay script

Your work is assessed against the UCA grading descriptors: Experimentation & innovation · Research & analysis · Engaging with practice · Realisation & communication · Personal & professional connectivity.

6 · Getting help

Need	Where to go
A coding error	Use the four-step routine (Read · Locate · Print · Search), then post the error and what you tried in the Q&A forum, or bring it to the code clinic in the tutorial slot
Course or assessment questions	Q&A forum — if you have the question, others do too
One-to-one help	Office hours (times are posted in Announcements)
Your coding pair	Your first point of help in labs; pairs rotate every three weeks
Referencing and library	UCA Harvard referencing guide (link in 'Portfolio & Research Skills') and the college library
Personal, wellbeing or access needs	Speak to your lecturer privately, or contact Student Support Services

7· Ground rules

In class and online

- Arrive on time with your laptop charged and the week's videos watched.
- Be respectful in discussion: evidence over opinion; challenge ideas, not people.
- Help your pair - explaining code is one of the best ways to learn it.
- Keep the Q&A forum professional and on topic.

Academic integrity and AI tools

- You **may** use AI assistants to explain an error, suggest search keywords or check grammar and you must declare that use in your R&D document.
- You **may not** submit writing or code produced by AI as your own, invent sources or data, or submit code you cannot explain line by line.
- Reference every source using the **UCA Harvard** style. AI chatbots can invent references that look real and always check that a source exists and read it yourself.
- Use only data, images and sound you have the right to use. Get consent before photographing or recording people or their work.

8 · Key words you will meet in Week 1

Term	Plain-English meaning
Artificial intelligence (AI)	Computer systems doing tasks we would call intelligent if a person did them
Machine learning (ML)	AI that learns patterns from examples instead of following hand-written rules

Term	Plain-English meaning
Deep learning	Machine learning using neural networks with many layers, good with images, sound and text
Model	What a machine-learning system learns: the 'recipe' it uses to make predictions
Dataset	The collection of examples a model learns from
Python	The programming language we use; popular for AI and readable for beginners
Notebook (.ipynb)	A file that mixes code, output and notes; runs in Google Colab or JupyterLab
Cell	One block of a notebook; run it with Shift + Enter
Variable	A labelled box that stores a value, e.g. artist = "Vera"
Loop	Code that repeats, e.g. for i in range(4):

9· Frequently asked questions

I have never coded. Will I be able to keep up?

Yes. Weeks 1–4 teach Python from the beginning through creative projects, every lab has a Core track, and you work in pairs. What matters most is attending, doing the weekly practice, and asking for help early.

Do I need a powerful laptop?

No. Heavy training runs in Google Colab, which provides free GPUs in the browser. Any laptop that can run a modern web browser is enough for most of the module.

Do I need to be good at maths?

No. We explain ideas visually and with analogies. You will use numbers, but the libraries do heavy maths.

Can my project be about my own discipline?

Yes, that is the point. The strongest projects connect AI to a real question in your own practice.

What if I miss a session?

Slides, notebooks and handouts are on Moodle each week. Work through them, post questions in the Q&A forum, and let your lecturer know.